

T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content $\text{SM}+S_1+S_2$. The report combines the renormalisable RGE running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F . Throughout, $16\pi^2\mu dX/d\mu=\beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Renormalisable EFT1 couplings

$\beta_{g_Y}^{(1)}$

$$16\pi^2\mu\frac{dg_Y}{d\mu}=\beta_{g_Y}^{(1)}$$

Column Representative term structure	
T1	$\frac{43g_Y^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{43g_Y^3}{6}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{43g_Y^3}{6}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{47g_Y^3}{6}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{47g_Y^3}{6}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{47g_Y^3}{6}$

$$\beta_{g_2}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{17g_2^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17g_2^3}{6}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{17g_2^3}{6}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5g_2^3}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{5g_2^3}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11g_2^3}{6}$

$$\beta_{g_3}^{(1)}$$

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column Representative term structure

T1	$-7g_3^3$
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7g_3^3$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-7g_3^3$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$

$\beta_{Y_u}^{(1)}$

$16\pi^2\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$
T2	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$
T3	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_u)_{ij}$	$+3\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_u)_{ij}$

Terms 2 of 3

Column	Representative term structure	
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	
T3	$-8g_3^2\ (Y_u)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Column	Representative term structure
T1	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

$16\pi^2\,\mu\,\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$
T3	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$

$\beta_{m_{S_1}^2}^{(1)}$

$16\pi^2\mu\frac{dm_{S_1}^2}{d\mu}=\beta_{m_{S_1}^2}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$-\frac{9g_2^2m_{S_1}^2}{2}$
T2	$-\frac{3g_Y^2m_{S_1}^2}{2}$
T3	$+4\lambda_{12}m_{S_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_{S_1}^2}{2}$	$-\frac{3g_Y^2m_{S_1}^2}{2}$	$+4\lambda_{12}m_{S_2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2m_{S_1}^2}{2}$	$-\frac{3g_Y^2m_{S_1}^2}{2}$	$+4\lambda_{12}m_{S_2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$6\lambda_{12}m_{S_2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+2\lambda_{12}m_{S_2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure	
T1	$+6\lambda_{S_1} m_{S_1}^2$	
T2	$-12g_2^2 m_{S_1}^2$	
T3	$-6g_Y^2 m_{S_1}^2$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{S_1} m_{S_1}^2$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{S_1} m_{S_1}^2$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{S_1} m_{S_1}^2$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-12g_2^2 m_{S_1}^2$	$-6g_Y^2 m_{S_1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-12g_2^2 m_{S_1}^2$	—

Terms 3 of 3

Column	Representative term structure
T1	$+m_{S_1}^2(8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$
T2	$+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+m_{S_1}^2(8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_{S_1}^2(8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$	$+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{m^2_{S_2}}^{(1)}$

$16\pi^2\mu\frac{dm^2_{S_2}}{d\mu}=\beta_{m^2_{S_2}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$-\frac{9g_2^2m_{S_2}^2}{2}$
T2	$-\frac{3g_Y^2m_{S_2}^2}{2}$
T3	$+4\lambda_{12}m_{S_1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_{S_2}^2}{2}$	$-\frac{3g_Y^2m_{S_2}^2}{2}$	$+4\lambda_{12}m_{S_1}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2m_{S_2}^2}{2}$	$-\frac{3g_Y^2m_{S_2}^2}{2}$	$+4\lambda_{12}m_{S_1}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_{12}m_{S_1}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$6\lambda_{12}m_{S_1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$+6\lambda_{S_2} m_{S_2}^2$
T2	$-12g_2^2 m_{S_2}^2$
T3	$-6g_Y^2 m_{S_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{S_2} m_{S_2}^2$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{S_2} m_{S_2}^2$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-12g_2^2 m_{S_2}^2$	$-6g_Y^2 m_{S_2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{S_2} m_{S_2}^2$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-12g_2^2 m_{S_2}^2$	$-6g_Y^2 m_{S_2}^2$

Terms 3 of 3

Column	Representative term structure
T1	$+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$
T2	$+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	$+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_H}^{(1)}$

$16\pi^2\mu\frac{d\lambda_H}{d\mu}=\beta_{\lambda_H}^{(1)}$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$
T2	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$
T3	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_H(12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\text{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\text{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$

Terms 2 of 7

Column	Representative term structure
T1	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$
T2	$+8 \lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\left(\lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\right)^{*}$
T3	$+4 \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\left(\lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+8 \lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\left(\lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\right)^{*}$	$+4 \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\left(\lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\right)^{*}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+8 \lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\left(\lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\right)^{*}$	$+4 \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\left(\lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}\right)^{*}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	—	—

Terms 3 of 7

Column	Representative term structure
T1	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$
T2	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{T3} \left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$

Terms 4 of 7

Column	Representative term structure
T1	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$
T2	$-9g_2^2\lambda_H$
T3	$+\frac{3g_Y^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$	$-9g_2^2\lambda_H$	$+\frac{3g_Y^4}{4}$

Terms 5 of 7

Column	Representative term structure
T1	$+12\lambda_H^2$
T2	$+4\lambda_{H1}^2$
T3	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+2\lambda_{H1}^2$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	—

Terms 6 of 7

Column	Representative term structure
T1	$+4\lambda_{H2}^2$
T2	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3	$+3\lambda_{H2}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{H2}^2$	—	$+3\lambda_{H2}^{\text{Adj}^2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{H2}^2$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{H2}^2$	—	$+3\lambda_{H2}^{\text{Adj}^2}$

Column	Representative term structure
T1	$+3\lambda_{H1}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$

$\beta_{\lambda_{S_1}}^{(1)}$

$_{16\pi^2}\mu\frac{d\lambda_{S_1}}{d\mu}=\beta_{\lambda_{S_1}}^{(1)}$

Terms 1 of 8

Column	Representative term structure
T1	$8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$
T2	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$
T3	$+24\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\left(\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$	$+24\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\left(\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$+8\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$	$+24\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\left(\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 8

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$
T3	$-9g_2^2 \lambda_{S_1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$	$-9g_2^2 \lambda_{S_1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$	$-9g_2^2 \lambda_{S_1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 8

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{4}$
T2	$+4\lambda_{12}^2$
T3	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$6\lambda_{12}^2$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+2\lambda_{12}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^2$	—

Terms 4 of 8

Column	Representative term structure
T1	$+4\lambda_{H1}^2$
T2	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$
T3	$+12\lambda_{S1}^2$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$	$+12\lambda_{S1}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H1}^2$	$+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$	$+12\lambda_{S1}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	—	$+10\lambda_{S1}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{H1}^2$	—	$+14\lambda_{S1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}^2$	—	$+14\lambda_{S1}^2$

Terms 5 of 8

Column	Representative term structure
T1	$21g_2^4$
T2	$+g_Y^2\left(12g_2^2-12\lambda_{S_1}-6\lambda_{S_1}^{\text{Adj}}\right)$
T3	$+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$21g_2^4$	$+g_Y^2\left(12g_2^2-12\lambda_{S_1}-6\lambda_{S_1}^{\text{Adj}}\right)$	$+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$21g_2^4$	—	$+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$

Terms 6 of 8

Column	Representative term structure
T1	$+12g_Y^4$
T2	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$
T3	$+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+12g_Y^4$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$	$+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$	$+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$

Terms 7 of 8

Column	Representative term structure
T1	$+26\lambda_{S_1}^{\text{Adj}^2}$
T2	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3	$+6\lambda_{12}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+6\lambda_{12}^{\text{Adj}^2}$

Column	Representative term structure
T1	$+6\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{\text{Cross}^2}$

$\beta_{\lambda_{S_2}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$

Terms 1 of 8

Column	Representative term structure
T1	$8\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$
T2	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$
T3	$+24\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$8\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$	$+24\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$8\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$	$+24\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 8

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$
T3	$-9g_2^2 \lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$	$-9g_2^2 \lambda_{S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9g_2^4}{4}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$	$-9g_2^2 \lambda_{S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 8

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{4}$
T2	$+4\lambda_{12}^2$
T3	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{3g_Y^4}{4}$	$+4\lambda_{12}^2$	$+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+2\lambda_{12}^2$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$6\lambda_{12}^2$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^2$	—

Terms 4 of 8

Column	Representative term structure
T1	$+4\lambda_{H2}^2$
T2	$+\frac{\lambda_{H2}^{\text{Adj}2}}{4}$
T3	$+12\lambda_{S2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{\text{Adj}2}}{4}$	$+12\lambda_{S_2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H2}^2$	$+\frac{\lambda_{H2}^{\text{Adj}2}}{4}$	$+12\lambda_{S_2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+14\lambda_{S_2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+10\lambda_{S_2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+14\lambda_{S_2}^2$

Terms 5 of 8

Column	Representative term structure
T1	$21g_2^4$
T2	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$
T3	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$21g_2^4$	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$21g_2^4$	$+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$

Terms 6 of 8

Column	Representative term structure
T1	$+12g_Y{}^4$
T2	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$
T3	$+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12g_Y{}^4$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12g_Y{}^4$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$

Terms 7 of 8

Column	Representative term structure
T1	$+26\lambda_{S_2}^{\text{Adj}^2}$
T2	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3	$+6\lambda_{12}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+6\lambda_{12}^{\text{Adj}^2}$

Column	Representative term structure
T1	$+6\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{\text{Cross}^2}$

$\beta_{\lambda_{H1}}^{(1)}$

$16\pi^2\,\mu\,\frac{d\lambda_{H1}}{d\mu}=\beta_{\lambda_{H1}}^{(1)}$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$
T3	$+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 7

Column	Representative term structure
T1	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^*)$
T2	$+ 6 \lambda_{T3} \left(\lambda_{T3} \right)^*$
T3	$+ \frac{9 g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)+6\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)+6\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+\frac{9g_2^4}{4}$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+12\lambda_{T3}\left(\lambda_{T3}\right)^*$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+4\lambda_{T3}\left(\lambda_{T3}\right)^*$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{T3}\left(\lambda_{T3}\right)^*$	—

Terms 3 of 7

Column	Representative term structure
T1	$-9g_2^2\lambda_{H1}$
T2	$+\frac{3g_Y^4}{4}$
T3	$-3g_Y^2\lambda_{H1}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{H1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 7

Column	Representative term structure
T1	$+4\lambda_{12}\lambda_{H2}$
T2	$+6\lambda_H\lambda_{H1}$
T3	$+4\lambda_{H1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$

Terms 5 of 7

Column	Representative term structure
T1	$+6\lambda_{H1}\lambda_{S1}$
T2	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3	$-\frac{9g_2^2\lambda_{H1}}{2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{H1}\lambda_{S1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{H1}\lambda_{S1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}\lambda_{S1}$	—	$-\frac{9g_2^2\lambda_{H1}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+8\lambda_{H1}\lambda_{S1}$	—	$-\frac{33g_2^2\lambda_{H1}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H1}\lambda_{S1}$	—	$-\frac{33g_2^2\lambda_{H1}}{2}$

Terms 6 of 7

Column	Representative term structure
T1	$-\frac{3g_Y^2\lambda_{H1}}{2}$
T2	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1^{\text{Adj}}})$
T3	$+6g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{15g_Y^2\lambda_{H1}}{2}$	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1^{\text{Adj}}})$	$+6g_2^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H1}}{2}$	$\lambda_{H1}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1^{\text{Adj}}})$	$+6g_2^4$

Terms 7 of 7

Column	Representative term structure
T1	$+3g_Y{}^4$
T2	$+6\lambda_{H1}^{\text{Adj}^2}$
T3	$+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3g_Y{}^4$	$+6\lambda_{H1}^{\text{Adj}^2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{H1}^{\text{Adj}^2}$	$+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_{H2}}^{(1)}$

$16\pi^2\,\mu\,\frac{d\lambda_{H2}}{d\mu}=\beta_{\lambda_{H2}}^{(1)}$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_{H2}(6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$
T2	$+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H2}(6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$	$+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H2}(6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$	$+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{H2}(6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 7

Column	Representative term structure
T1	$+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$
T2	$+6\lambda_{T3} \left(\lambda_{T3} \right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	$+6\lambda_{T3} \left(\lambda_{T3} \right)^*$	$+\frac{9g_2^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	$+6\lambda_{T3} \left(\lambda_{T3} \right)^*$	$+\frac{9g_2^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{T3} \left(\lambda_{T3} \right)^*$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+12\lambda_{T3} \left(\lambda_{T3} \right)^*$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{T3} \left(\lambda_{T3} \right)^*$	—

Terms 3 of 7

Column	Representative term structure
T1	$-9g_2^2\lambda_{H2}$
T2	$+\frac{3g_Y^4}{4}$
T3	$-3g_Y^2\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H2}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{H2}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 7

Column	Representative term structure
T1	$+4\lambda_{12}\lambda_{H1}$
T2	$+6\lambda_H\lambda_{H2}$
T3	$+4\lambda_{H2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$

Terms 5 of 7

Column	Representative term structure
T1	$+6\lambda_{H2}\lambda_{S2}$
T2	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3	$\lambda_{H2}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{S2}^{\text{Adj}})$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{H2}\lambda_{S2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{H2}\lambda_{S2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{S2}$	—	$\lambda_{H2}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{S2}^{\text{Adj}})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}\lambda_{S2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{S2}$	—	$\lambda_{H2}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{S2}^{\text{Adj}})$

Terms 6 of 7

Column	Representative term structure
T1	$+6g_2^4$
T2	$-\frac{33g_2^2\lambda_{H2}}{2}$
T3	$+3g_Y^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{9g_2^2\lambda_{H2}}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$+3g_Y^4$

Terms 7 of 7

Column	Representative term structure
T1	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T2	$+6\lambda_{H2}^{\text{Adj}^2}$
T3	$+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3g_Y^2\lambda_{H2}}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	$+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$

$\beta_{\lambda_{12}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 9

Column	Representative term structure
T1	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$
T3	$+9\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$	$+9\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$	$+9\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 9

Column	Representative term structure
T1	$+9\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right)$
T2	$+3\lambda_{T3} \left(\lambda_{T3} \right)^*$
T3	$+ \frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+9\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) + 3\lambda_{T3} \left(\lambda_{T3} \right)^*$	$+ \frac{9g_2^4}{4}$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+9\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) + 3\lambda_{T3} \left(\lambda_{T3} \right)^*$	$+ \frac{9g_2^4}{4}$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$4\lambda_{T3} \left(\lambda_{T3} \right)^*$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$4\lambda_{T3} \left(\lambda_{T3} \right)^*$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 9

Column	Representative term structure
T1	$-9g_2^2\lambda_{12}$
T2	$+\frac{3g_Y^4}{4}$
T3	$-3g_Y^2\lambda_{12}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{12}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{12}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{12}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{12}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12g_2^2\lambda_{12}$	—	$-6g_Y^2\lambda_{12}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12g_2^2\lambda_{12}$	—	$-6g_Y^2\lambda_{12}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 9

Column	Representative term structure
T1	$+4\lambda_{12}^2$
T2	$+6\lambda_{12}\lambda_{S_1}$
T3	$+6\lambda_{12}\lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}^2$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}^2$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}^2$	$+4\lambda_{12}\lambda_{S_1}$	$+8\lambda_{12}\lambda_{S_2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{12}^2$	$+8\lambda_{12}\lambda_{S_1}$	$+4\lambda_{12}\lambda_{S_2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}^2$	—	—

Terms 5 of 9

Column	Representative term structure
T1	$+ \frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T2	$+4\lambda_{H1}\lambda_{H2}$
T3	$+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{3\lambda_{12}^{\text{Adj}^2}}{4}$	$+4\lambda_{H1}\lambda_{H2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{3\lambda_{12}^{\text{Adj}^2}}{4}$	$+4\lambda_{H1}\lambda_{H2}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{H1}\lambda_{H2}$	$+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+4\lambda_{H1}\lambda_{H2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{H1}\lambda_{H2}$	—

Terms 6 of 9

Column	Representative term structure
T1	$+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$
T2	$\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3	$+16g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$	$+16g_2^4$

Terms 7 of 9

Column	Representative term structure
T1	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$
T2	$+g_Y^2(-6\lambda_{12}-4\lambda_{12}^{\text{Cross}})$
T3	$+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$	$+g_Y^2(-6\lambda_{12}-4\lambda_{12}^{\text{Cross}})$	$+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$

Terms 8 of 9

Column	Representative term structure	
T1	$+ \lambda_{12} \left(\frac{16 \lambda_{12}^{\text{Cross}}}{3} + 4 \lambda_{S_1}^{\text{Adj}} + 4 \lambda_{S_2}^{\text{Adj}} \right)$	
T2	$+ \lambda_{S_2} \left(8 \lambda_{12} + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$	
T3	$+ 16 \lambda_{12}^{\text{Adj}^2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \lambda_{12} \left(\frac{16 \lambda_{12}^{\text{Cross}}}{3} + 4 \lambda_{S_1}^{\text{Adj}} + 4 \lambda_{S_2}^{\text{Adj}} \right)$	$+ \lambda_{S_2} \left(8 \lambda_{12} + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$	$+ 16 \lambda_{12}^{\text{Adj}^2}$

Terms 9 of 9

Column	Representative term structure	
T1	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$	
T2	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$	
T3	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$

$\beta_{\lambda T3}^{(1)}$

$16\pi^2\,\mu\,\frac{d\lambda_{T3}}{d\mu}=\beta_{\lambda T3}^{(1)}$

Terms 1 of 5

Column	Representative term structure	
T1	λ_{T3}	$\left(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T2	$-4\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}$	$\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$
T3	$+4\lambda_{S_1S_1^{\dagger}2S_2^{\dagger}}$	$\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{T3}\left(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$	$-4\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$	$+4\lambda_{S_1S_1^{\dagger}2S_2^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{T3}\left(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^{\dagger}\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^{\dagger}\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^{\dagger}\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$	$-4\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$	$+4\lambda_{S_1S_1^{\dagger}2S_2^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T2	$-9g_2^2\lambda_{T3}$
T3	$-3g_Y^2\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	$-9g_2^2\lambda_{T3}$	$-3g_Y^2\lambda_{T3}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	$-9g_2^2\lambda_{T3}$	$-3g_Y^2\lambda_{T3}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$+2\lambda_{12}\lambda_{T3}$
T2	$+2\lambda_H\lambda_{T3}$
T3	$+4\lambda_{H1}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_{H2}\lambda_{T3}$
T2	$\lambda_{T3}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{H2}^{\text{Adj}})$
T3	$-\frac{21g_2^2\lambda_{T3}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	$\lambda_{T3}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{H2}^{\text{Adj}})$	$-\frac{21g_2^2\lambda_{T3}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	$-\frac{21g_2^2\lambda_{T3}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	$-\frac{33g_2^2\lambda_{T3}}{2}$

Terms 5 of 5

Column	Representative term structure
T1	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T2	$\lambda_{T3}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{H1}^{\text{Adj}})$
T3	$\lambda_{T3}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-\frac{4\lambda_{12}^{\text{Adj}}}{3}-2\lambda_{12}^{\text{Cross}}-2\lambda_{H1}^{\text{Adj}}-\frac{4\lambda_{H2}^{\text{Adj}}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$	$\lambda_{T3}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{H1}^{\text{Adj}})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$	—	$\lambda_{T3}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-\frac{4\lambda_{12}^{\text{Adj}}}{3}-2\lambda_{12}^{\text{Cross}}-2\lambda_{H1}^{\text{Adj}}-\frac{4\lambda_{H2}^{\text{Adj}}}{3}\right)$

$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{64}{3}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$
T2	$+\frac{4}{3}\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}\left(5\left(\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*\right)$
T3	$+\frac{4}{3}\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\left(\left(\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}\right)^*+5\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{64}{3}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$	$+\frac{4}{3}\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}\left(5\left(\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*\right)+\frac{4}{3}\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\left(\left(\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}\right)^*+5\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*\right)$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{64}{3}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$	$+\frac{4}{3}\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}\left(5\left(\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*\right)+\frac{4}{3}\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\left(\left(\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}\right)^*+5\left(\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}\right)^*\right)$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T2	$+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$
T3	$-9g_2^2\lambda_{12}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{12}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$	$-9g_2^2\lambda_{12}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2}{9}\lambda_{T3}\left(\lambda_{T3}\right)^*$	—	$-24g_2^2\lambda_{12}^{\text{Adj}}$

Terms 3 of 5

Column	Representative term structure
T1	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T2	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$
T3	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$

Terms 4 of 5

Column	Representative term structure
T1	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T2	$-6g_Y{}^2\lambda_{12}^{\text{Adj}}$
T3	$+\frac{16\lambda_{12}^{\text{Adj}}\lambda_{12}^{\text{Cross}}}{3}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-6g_Y{}^2\lambda_{12}^{\text{Adj}}$	$+\frac{16\lambda_{12}^{\text{Adj}}\lambda_{12}^{\text{Cross}}}{3}$

Terms 5 of 5

Column	Representative term structure
T1	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$
T2	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$
T3	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$
T2	$+\frac{33g_2^4}{4}$
T3	$+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$

Terms 2 of 5

Column	Representative term structure
T1	$+g_Y{}^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$
T2	$+\lambda_{12}{}^2$
T3	$+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$	$+\lambda_{12}{}^2$	$+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$

Terms 3 of 5

Column	Representative term structure
T1	$+\lambda_{12}(8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}})$
T2	$+\lambda_{S_2}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$
T3	$+4\lambda_{12}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{12}(8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}})$	$+\lambda_{S_2}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$	$+4\lambda_{12}^{\text{Adj}^2}$

Terms 4 of 5

Column	Representative term structure
T1	$+11\lambda_{12}^{\text{Cross}^2}$
T2	$-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$
T3	$-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+11\lambda_{12}^{\text{Cross}^2}$	$-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$	$-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$

Terms 5 of 5

Column	Representative term structure							
T1	$+ \lambda_{H1} \lambda_{H2}$							

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \lambda_{H1} \lambda_{H2}$

$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$

$$16\pi^2\,\mu\frac{d\lambda_{H1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{H1}^{\text{Adj}}(6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$
T2	$+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$
T3	$-\frac{4}{3}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}^{\text{Adj}}(6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$	$+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$-\frac{4}{3}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H1}^{\text{Adj}}(6\,\text{Tr}\big(Y_d\,\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,\,(Y_u)^\dagger\big))$	$+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$	$-\frac{4}{3}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure	
T1	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$	
T2	$+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	
T3	$+g_Y^2\left(-6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(-6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	$+g_Y^2\left(-6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$	—

Terms 3 of 6

Column	Representative term structure
T1	$-9g_2^2\lambda_{H1}^{\text{Adj}}$
T2	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_H\lambda_{H1}^{\text{Adj}}$

Terms 4 of 6

Column	Representative term structure
T1	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$
T2	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$
T3	$\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S1}^{\text{Adj}})$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	$\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$	$\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S1}^{\text{Adj}})$

Terms 5 of 6

Column	Representative term structure	
T1	$+g_Y^2\left(-4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$	
T2	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$	
T3	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_Y^2\left(-4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$	$-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$

Column	Representative term structure
T1	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$

$_{16\pi^2}\mu\frac{d\lambda_{H2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$

Terms 1 of 6

Column	Representative term	structure
T1	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	
T2	$-4\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	
T3	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$-4\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$-4\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure	
T1	$+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	
T2	$+16\lambda_{T3} \left(\lambda_{T3} \right)^*$	
T3	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H2}^{\text{Adj}} \right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	$+16\lambda_{T3} \left(\lambda_{T3} \right)^*$	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H2}^{\text{Adj}} \right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	$+16\lambda_{T3} \left(\lambda_{T3} \right)^*$	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H2}^{\text{Adj}} \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 6

Column	Representative term structure
T1	$-9g_2^2\lambda_{H2}^{\text{Adj}}$
T2	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$

Terms 4 of 6

Column	Representative term structure
T1	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T2	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$
T3	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$

Terms 5 of 6

Column	Representative term structure
T1	$+\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$
T2	$+g_Y{}^2\left(4g_2{}^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$
T3	$-\frac{33g_2{}^2\lambda_{H2}^{\text{Adj}}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+g_Y{}^2\left(4g_2{}^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$	$-\frac{33g_2{}^2\lambda_{H2}^{\text{Adj}}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{8}{9}\lambda_{T3}\left(\lambda_{T3}\right)^{*}$	$+g_Y{}^2\left(4g_2{}^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$	$-\frac{33g_2{}^2\lambda_{H2}^{\text{Adj}}}{2}$

Column	Representative term structure
T1	$+12\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$

$\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$
T3	$+2\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}(\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$	$+2\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}(\lambda_{T3})^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$	$+2\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}(\lambda_{T3})^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T2	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure	
T1	$+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	
T2	$+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$	
T3	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S_1}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S_1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S_1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Column	Representative term structure
T1	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$$\beta_{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T2	$+6\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3	$-6\left(\lambda_{H^\dagger H^\dagger S_2S_2}\right)^*\left(\lambda_{T3}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+6\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_{H^\dagger H^\dagger S_2S_2}\right)^*\left(\lambda_{T3}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+6\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_{H^\dagger H^\dagger S_2S_2}\right)^*\left(\lambda_{T3}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure	
T1	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T2	$-3g_Y^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T3	$+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-3g_Y^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-3g_Y^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$+\lambda_{H1}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger})$
T3	$+\lambda_{H1}^{\text{Adj}}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+\lambda_{H1}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger})$	$+\lambda_{H1}^{\text{Adj}}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+\lambda_{H1}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger})$	$+\lambda_{H1}^{\text{Adj}}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_{H2}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}})$
T2	$+\lambda_{H2}^{\text{Adj}}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}{4}\right)$
T3	$-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{H2}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}})$	$+\lambda_{H2}^{\text{Adj}}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}{4}\right)$	$-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{T3}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{H2}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}})$	$+\lambda_{H2}^{\text{Adj}}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}{4}\right)$	$-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{T3}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{H^\dagger H^\dagger S_2 S_2}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{H^\dagger H^\dagger S_2 S_2}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$
T3	$+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H^\dagger H^\dagger S_2 S_2}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$	$+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H^\dagger H^\dagger S_2 S_2}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$	$+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$
T2	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$	$-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$
T2	$+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$
T3	$+2\lambda_{H^\dagger H^\dagger S_2 S_2}^{\lambda S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+2\lambda_{H^\dagger H^\dagger S_2 S_2}^{\lambda S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$	$+2\lambda_{H^\dagger H^\dagger S_2 S_2}^{\lambda S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{3g_2^4}{4}$
T2	$+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_1}}{2}-12\lambda_{S_1}^{\text{Adj}}\right)$
T3	$+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3g_2^4}{4}$	$+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_1}}{2}-12\lambda_{S_1}^{\text{Adj}}\right)$	$+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$

Terms 2 of 5

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{2}$
T2	$+ \frac{\lambda_{12}^2}{4}$
T3	$+ \frac{\lambda_{H1}^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3g_Y^4}{2}$	$+ \frac{\lambda_{12}^2}{4}$	$+ \frac{\lambda_{H1}^2}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+ \frac{\lambda_{H1}^2}{2}$

Terms 3 of 5

Column	Representative term structure
T1	$+3\lambda_{H1}^{\text{Adj}2}$
T2	$+\frac{7\lambda_{S1}^2}{4}$
T3	$+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$

Terms 4 of 5

Column	Representative term structure
T1	$+22\lambda_{S_1}^{\text{Adj}^2}$
T2	$+\frac{3\lambda_{12}^2}{4}$
T3	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+22\lambda_{S_1}^{\text{Adj}^2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_1}^{\text{Adj}^2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$

Terms 5 of 5

Column	Representative term structure
T1	$+12\lambda_{12}^{\text{Adj}^2}$
T2	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}^2}$	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

$$\beta_{\lambda_{S_1 S_1^\dagger S_2^\dagger}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_1 S_1^\dagger S_2^\dagger}}{d\mu}=\beta_{\lambda_{S_1 S_1^\dagger S_2^\dagger}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\lambda_{S_1^\dagger S_2^\dagger S_2^\dagger}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}\right)^*+5\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}\right)^*)$
T2	$-9g_2^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$
T3	$-3g_Y^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_{S_1^\dagger S_2^\dagger S_2^\dagger}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}\right)^*+5\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}\right)^*)$	$-9g_2^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$	$-3g_Y^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$4\lambda_{S_1^\dagger S_2^\dagger S_2^\dagger}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}\right)^*+5\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}\right)^*)$	$-9g_2^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$	$-3g_Y^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$+ \lambda_{12} (6 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$
T2	$+ \lambda_{12}^{\text{Adj}} \left(\frac{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{5 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6} \right)$
T3	$+ 2 \lambda_{H1} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_{12} (6 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$	$+ \lambda_{12}^{\text{Adj}} \left(\frac{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{5 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6} \right)$	$+ 2 \lambda_{H1} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{12} (6 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$	$+ \lambda_{12}^{\text{Adj}} \left(\frac{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{5 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6} \right)$	$+ 2 \lambda_{H1} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure	
T1	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	
T2	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda_{T3}}$	
T3	$+12\lambda_{S_1}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda_{T3}}$	$+12\lambda_{S_1}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda_{T3}}$	$+12\lambda_{S_1}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$$\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}{d\mu}=\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$
T2	$-9g_2^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$
T3	$-3g_Y^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$	$-9g_2^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$	$-3g_Y^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$	$-9g_2^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$	$-3g_Y^2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda_{12}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$
T2	$+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}}{3}$
T3	$+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{12}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$	$+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}}{3}$	$+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}^2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{12}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$	$+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}}{3}$	$+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}^2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_{S_1} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$
T2	$+2\lambda_{S_1^\dagger 2 S_2^\dagger 2} \lambda_{S_2}$
T3	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{S_1} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$	$+2\lambda_{S_1^\dagger 2 S_2^\dagger 2} \lambda_{S_2}$	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{S_1} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$	$+2\lambda_{S_1^\dagger 2 S_2^\dagger 2} \lambda_{S_2}$	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}$
T2	$+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}$	$+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}$	$+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$$\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger{}^2}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{S_1^\dagger S_2 S_2^\dagger{}^2}}{d\mu}=\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger{}^2}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$
T2	$+4\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger{}^2}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$
T3	$-9g_2^2\lambda_{S_1^\dagger S_2 S_2^\dagger{}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$	$+4\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger{}^2}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$	$-9g_2^2\lambda_{S_1^\dagger S_2 S_2^\dagger{}^2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$	$+4\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger{}^2}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$	$-9g_2^2\lambda_{S_1^\dagger S_2 S_2^\dagger{}^2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$-3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger^2}$
T2	$+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger^2}+6\lambda_{S_1 S_1^\dagger^2 S_2^\dagger})$
T3	$+\lambda_{12}^{\text{Adj}}\left(\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6}+\frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger^2}$	$+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger^2}+6\lambda_{S_1 S_1^\dagger^2 S_2^\dagger})$	$+\lambda_{12}^{\text{Adj}}\left(\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6}+\frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger^2}$	$+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger^2}+6\lambda_{S_1 S_1^\dagger^2 S_2^\dagger})$	$+\lambda_{12}^{\text{Adj}}\left(\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6}+\frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$+12\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$+12\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$+12\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{3g_2^4}{4}$
T2	$+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$
T3	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3g_2^4}{4}$	$+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3g_2^4}{4}$	$+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$

Terms 2 of 5

Column	Representative term structure
T1	$+\frac{3g_Y^4}{2}$
T2	$+\frac{\lambda_{12}^2}{4}$
T3	$+\frac{\lambda_{H2}^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_Y^4}{2}$	$+\frac{\lambda_{12}^2}{4}$	$+\frac{\lambda_{H2}^2}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_Y^4}{2}$	—	$+\frac{\lambda_{H2}^2}{2}$

Terms 3 of 5

Column	Representative term structure
T1	$+3\lambda_{H2}^{\text{Adj}^2}$
T2	$+\frac{7\lambda_{S2}^2}{4}$
T3	$+13\lambda_{S2}\lambda_{S2}^{\text{Adj}}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$	$+13\lambda_{S2}\lambda_{S2}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$	$+13\lambda_{S2}\lambda_{S2}^{\text{Adj}}$

Terms 4 of 5

Column	Representative term structure
T1	$+22\lambda_{S_2}^{\text{Adj}^2}$
T2	$+\frac{3\lambda_{12}^2}{4}$
T3	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$

Terms 5 of 5

Column	Representative term structure
T1	$+12\lambda_{12}^{\text{Adj}^2}$
T2	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}^2}$	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab} . Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$$\beta_{C_{1133}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1133}}{d\mu}=\beta_{C_{1133}}^{(1)}$$

Column	Representative term structure
T 1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T 2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T 1	T 2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{4lambdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{1134}}^{(1)}$

$16\pi^2\mu\frac{dC_{1134}}{d\mu}=\beta_{C_{1134}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{i\lambda bdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{4i\lambda bdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{2i\lambda bdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{1144}}^{(1)}$

$16\pi^2\,_{\mu}\frac{dC_{1144}}{d\mu}=\beta_{C_{1144}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{4lambdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\frac{2lambdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{1155}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1155}}{d\mu}=\beta_{C_{1155}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{ye\overline{y_1}^2\overline{y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\overline{S_1}Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$-\frac{\sqrt{3}\lambda\overline{S_1}Inv1\overline{y_1}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{1156}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1156}}{d\mu}=\beta_{C_{1156}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{i\lambda S1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{i\lambda S1Inv1\overline{y_1}^2}{2MF}$	$\frac{3ig_2^2\overline{y_1}^2}{4MF}$	$-\frac{iy_e\overline{y_1}^2\overline{y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{1157}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1157}}{d\mu}=\beta_{C_{1157}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Column	Representative term structure
T1	$\frac{3^4 ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^4 ye\overline{y_1y_2ye}}{12MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{1158}}^{(1)}$

$16\pi^2\mu\frac{dC_{1158}}{d\mu}=\beta_{C_{1158}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Column	Representative term structure
T1	$\frac{3^4 i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^4 i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{11511}}^{(1)}$

$16\pi^2\mu\frac{dC_{11511}}{d\mu}=\beta_{C_{11511}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\overline{lambda12Inv2y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}\overline{lambda12Inv2y_1y_2}}{3MF}$	$\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{2\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{2MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{6MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T3	$\frac{3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3 M F}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3 M F}$	—	$-\frac{17 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{24 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda\mathrm{Inv1}\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}\lambda\mathrm{Inv1}\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{11512}}^{(1)}$

$16\pi^2\mu\frac{dC_{11512}}{d\mu}=\beta_{C_{11512}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inu2\overline{y_1y_2}}{MF}$
T3	$\frac{iy_e\overline{y_1y_2y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12Inu2\overline{y_1y_2}}{3MF}$	$\frac{iy_e\overline{y_1y_2y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{ilambda12Inv2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{2ilambda12Inv2y_1y_2}{3MF}$	$\frac{\sqrt{3}ig_2^2y_1y_2}{2MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{6MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y \overline{e}}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y \overline{e}}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3MF}$	$-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y \overline{e}}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y \overline{e}}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{11513}}^{(1)}$

$16\pi^2\mu\frac{dC_{11513}}{d\mu}=\beta_{C_{11513}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{2}g_1^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{66}\lambda 12Inv2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33\sqrt{2}g_2^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{2}g_1^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{66}\lambda 12Inv2\overline{y_1}y_2}{44MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}lambda12Inv1y_1y_2}{MF}$
T2	$\frac{\sqrt{2}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y}_1\overline{y}_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}lambda12Inv1y_1y_2}{4MF}$	$-\frac{\sqrt{2}ig_2^2y_1y_2}{4MF}$	$\frac{\sqrt{2}ye\overline{y}_1\overline{y}_2\overline{y}e}{8MF}$

$\beta_{C_{11514}}^{(1)}$

$16\pi^2\mu\frac{dC_{11514}}{d\mu}=\beta_{C_{11514}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{2}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{2}ig_1^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}g_2^2y_1y_2}{4MF}$	$-\frac{33\sqrt{2}ig_2^2y_1y_2}{16MF}$	$-\frac{3\sqrt{2}ig_1^2y_1y_2}{16MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{2}i\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2}i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{44 M F}$	$\frac{\sqrt{2}i\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{4 M F}$	$\frac{\sqrt{2}i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{1166}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1166}}{d\mu}=\beta_{C_{1166}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{1167}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1167}}{d\mu}=\beta_{C_{1167}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3}{3^4}\frac{ilambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3}{3^4}\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3}{3^4}\frac{ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3}{3^4}\frac{ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3}{3^4}\frac{ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3}{8MF}\frac{3}{3^4}\frac{ig_2^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Column	Representative term structure
T1	$\frac{3^4 i y e \overline{y_1 y_2 y e}}{12 M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^4 i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{1168}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1168}}{d\mu}=\beta_{C_{1168}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Column	Representative term structure
T1	$\frac{3^4 ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^4 ye\overline{y_1y_2ye}}{12MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{11611}}^{(1)}$

$$16\pi^2\mu\frac{dC_{11611}}{d\mu}=\beta_{C_{11611}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{iy_e\overline{y_1y_2y\bar{e}}}{MF}$
T3	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{iy_e\overline{y_1y_2y\bar{e}}}{4MF}$	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1y_2}}{3MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{3} i y e \overline{y_1 y_2} y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2 i \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3 M F}$	$-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{2 M F}$	$\frac{\sqrt{3} i y e \overline{y_1 y_2} y e}{6 M F}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye \overline{y_1} y_2 y \overline{e}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1} y_2}{MF}$
T3	$\frac{3^{\frac{3}{4}} i \lambda 12 Inv 1 \overline{y_1} y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} ye \overline{y_1} y_2 y \overline{e}}{12MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1} y_2}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1} y_2}{3MF}$	$\frac{3^{\frac{3}{4}} i \lambda 12 Inv 1 \overline{y_1} y_2}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{11612}}^{(1)}$

$16\pi^2\mu\frac{dC_{11612}}{d\mu}=\beta_{C_{11612}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\overline{lambda12Inv2y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}\overline{lambda12Inv2y_1y_2}}{3MF}$	$\frac{ye\overline{y_1y_2y\overline{e}}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\lambda 12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{2\lambda 12Inv2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{2MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{6MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T3	$\frac{3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3 M F}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3 M F}$	—	$-\frac{17 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{24 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda\mathrm{Inv1}\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}\lambda\mathrm{Inv1}\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{11613}}^{(1)}$

$16\pi^2\mu\frac{dC_{11613}}{d\mu}=\beta_{C_{11613}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{2} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{44 M F}$	$\frac{33\sqrt{2} i g_2^2 \overline{y_1 y_2}}{16 M F}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{11614}}^{(1)}$

$$16\pi^2\mu\frac{dC_{11614}}{d\mu}=\beta_{C_{11614}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{2}g_1^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{66}\lambda\text{Inv}2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33\sqrt{2}g_2^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{2}g_1^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{66}\lambda\text{Inv}2\overline{y_1}y_2}{44MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}lambda12Inv1y_1y_2}{MF}$
T2	$\frac{\sqrt{2}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}lambda12Inv1y_1y_2}{4MF}$	$-\frac{\sqrt{2}ig_2^2y_1y_2}{4MF}$	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{11711}}^{(1)}$

$16\pi^2\mu\frac{dC_{11711}}{d\mu}=\beta_{C_{11711}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$	$-\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{66}\text{lambda}12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}\text{lambda}12Inv2\overline{y_1y_2}}{44MF}$	$\frac{49\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{2}ye\overline{y_1y_2ye}}{8MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$

$\beta_{C_{11712}}^{(1)}$

$16\pi^2\mu\frac{dC_{11712}}{d\mu}=\beta_{C_{11712}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2\overline{y_1y_2}}{MF}$	$-\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{4MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$	$\frac{3\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{44MF}$	$\frac{49\sqrt{2}ig_2^2\overline{y_1y_2}}{16MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{11715}}^{(1)}$

$16\pi^2\mu\frac{dC_{11715}}{d\mu}=\beta_{C_{11715}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{2}i\lambda a12Inv1\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}ig_1^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{4MF}$	$-\frac{\sqrt{2}i\lambda a12Inv1\overline{y_1}y_2}{4MF}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1}y_2}{16MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{2} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{44 M F}$	$\frac{33\sqrt{2} i g_2^2 \overline{y_1 y_2}}{16 M F}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{11716}}^{(1)}$

$16\pi^2\mu\frac{dC_{11716}}{d\mu}=\beta_{C_{11716}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{2}g_1^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{66}\lambda12Inv2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33\sqrt{2}g_2^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{2}g_1^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{66}\lambda12Inv2\overline{y_1}y_2}{44MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}lambda12Inv1y_1y_2}{MF}$
T2	$\frac{\sqrt{2}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}lambda12Inv1y_1y_2}{4MF}$	$\frac{\sqrt{2}ig_2^2y_1y_2}{4MF}$	$\frac{\sqrt{2}ye\overline{y_1y_2ye}}{8MF}$

$\beta_{C_{11811}}^{(1)}$

$16\pi^2\mu\frac{dC_{11811}}{d\mu}=\beta_{C_{11811}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{49\sqrt{2}ig_2^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{2}ig_1^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{44MF}$	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{11812}}^{(1)}$

$16\pi^2\mu\frac{dC_{11812}}{d\mu}=\beta_{C_{11812}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$	$-\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{66}\text{lambda}12Inv2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}\text{lambda}12Inv2\overline{y_1}y_2}{44MF}$	$\frac{49\sqrt{2}g_2^2\overline{y_1}y_2}{16MF}$	$-\frac{\sqrt{2}ye\overline{y_1}y_2\overline{y}e}{8MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$

$\beta_{C_{11815}}^{(1)}$

$16\pi^2\mu\frac{dC_{11815}}{d\mu}=\beta_{C_{11815}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{66}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$	$\frac{3\sqrt{66}lambda12Inv2\overline{y_1y_2}}{44MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{2}ig_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{33\sqrt{2}g_2^2\overline{y_1}y_2}{16MF}$	$-\frac{\sqrt{2}ig_2^2\overline{y_1}y_2}{4MF}$	$-\frac{\sqrt{2}ye\overline{y_1}y_2\overline{y}e}{8MF}$

$\beta_{C_{11816}}^{(1)}$

$16\pi^2\mu\frac{dC_{11816}}{d\mu}=\beta_{C_{11816}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}i\lambda a12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}i\lambda a12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{2} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{44 M F}$	$\frac{33\sqrt{2} i g_2^2 \overline{y_1 y_2}}{16 M F}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{11911}}^{(1)}$

$16\pi^2\mu\frac{dC_{11911}}{d\mu}=\beta_{C_{11911}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Column	Representative term structure
T1	$\frac{3^4 i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3^4 i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{11912}}^{(1)}$

$16\pi^2\mu\frac{dC_{11912}}{d\mu}=\beta_{C_{11912}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^4 ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^4 ye\overline{y_1y_2ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{11913}}^{(1)}$

$$16\pi^2\mu\frac{dC_{11913}}{d\mu}=\beta_{C_{11913}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{2}g_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{2}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{ig_2^2y_1y_2}{MF}$	$-\frac{\sqrt{2}g_2^2y_1y_2}{4MF}$	$-\frac{49\sqrt{2}ig_2^2y_1y_2}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{2}ig_1^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{44MF}$	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{11914}}^{(1)}$

$16\pi^2\mu\frac{dC_{11914}}{d\mu}=\beta_{C_{11914}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$	$-\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{66}\text{lambda}12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}\text{lambda}12Inv2\overline{y_1y_2}}{44MF}$	$\frac{49\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{111011}}^{(1)}$

$_{16\pi^2} \mu \frac{dC_{111011}}{d\mu} = \beta_{C_{111011}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3}{3^4} \frac{g_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3}{3^4} \frac{lambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3}{3^4} \frac{g_1^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{11 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}} lambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^4 ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3^4 ye\overline{y_1y_2ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{111012}}^{(1)}$

$_{16\pi^2}^2\mu\frac{dC_{111012}}{d\mu}=\beta_{C_{111012}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3}{3^4}ig_2^2\overline{y_1y_2}$ MF
T2	$\frac{3}{3^4}ilambda12Inv1\overline{y_1y_2}$ MF
T3	$\frac{3}{3^4}ig_1^2\overline{y_1y_2}$ MF

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Column	Representative term structure
T1	$\frac{3^4 i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3^4 i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{C_{111013}}^{(1)}$

$_{16\pi^2}{}_{\mu}\frac{dC_{111013}}{d\mu}=\beta_{C_{111013}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2\overline{y_1y_2}}{MF}$	$-\frac{49\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{66}\lambda\text{Inv}2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}\lambda\text{Inv}1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{66}\lambda\text{Inv}2\overline{y_1y_2}}{44MF}$	$\frac{\sqrt{2}\lambda\text{Inv}1\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{111014}}^{(1)}$

$_{16\pi^2} \mu \frac{dC_{111014}}{d\mu} = \beta_{C_{111014}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2 y_1 y_2}{MF}$
T2	$\frac{\sqrt{2}g_2^2 y_1 y_2}{MF}$
T3	$\frac{\sqrt{2}ig_2^2 y_1 y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{ig_2^2 y_1 y_2}{MF}$	$-\frac{\sqrt{2}g_2^2 y_1 y_2}{4MF}$	$-\frac{49\sqrt{2}ig_2^2 y_1 y_2}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{2}ig_1^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{44MF}$	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{111111}}^{(1)}$

$_{16\pi^2}{}_{\mu}\frac{dC_{111111}}{d\mu}=\beta_{C_{111111}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{ye\overline{y_2}^2\overline{y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\overline{S_2}Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}\lambda\overline{S_2}Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{111112}}^{(1)}$

$16\pi^2\,\mu\frac{dC_{111112}}{d\mu}=\beta_{C_{111112}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{iy_e\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{ilambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{iy_e\overline{y_2}^2\overline{y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{111212}}^{(1)}$

$_{16\pi^2}^2\mu\frac{dC_{111212}}{d\mu}=\beta_{C_{111212}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_2}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_2}^2\overline{y\overline{e}}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{1213}}^{(1)}$

$16\pi^2\mu\frac{dC_{1213}}{d\mu}=\beta_{C_{1213}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{1214}}^{(1)}$

$16\pi^2\mu\frac{dC_{1214}}{d\mu}=\beta_{C_{1214}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\frac{ilambdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{1223}}^{(1)}$

$16\pi^2\mu\frac{dC_{1223}}{d\mu}=\beta_{C_{1223}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{i\lambda bdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{2i\lambda bdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\frac{i\lambda bdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{1224}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1224}}{d\mu}=\beta_{C_{1224}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{lambdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{1257}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1257}}{d\mu}=\beta_{C_{1257}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1}^2}{8MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{ye\overline{y_1}^2\overline{y_e}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\overline{S_1}Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{8MF}$	$-\frac{\sqrt{3}\lambda\overline{S_1}Inv1\overline{y_1}^2}{12MF}$	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{24MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{1258}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1258}}{d\mu}=\beta_{C_{1258}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iy e\overline{y_1}^2\overline{y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3ig_2^2\overline{y_1}^2}{8MF}$	$-\frac{iy e\overline{y_1}^2\overline{y e}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{24MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{1259}}^{(1)}$

$16\pi^2\mu\frac{dC_{1259}}{d\mu}=\beta_{C_{1259}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{ye\overline{y_1y_2y\overline{e}}}{8MF}$	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{6MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{12510}}^{(1)}$

$16\pi^2\mu\frac{dC_{12510}}{d\mu}=\beta_{C_{12510}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{iy_e\overline{y_1y_2y\bar{e}}}{MF}$
T3	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{iy_e\overline{y_1y_2y\bar{e}}}{8MF}$	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1y_2}}{6MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$\frac{i\lambda 12Inv2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{i\lambda 12Inv2y_1y_2}{3MF}$	$-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T3	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$$\beta_{C_{12511}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12511}}{d\mu}=\beta_{C_{12511}}^{(1)}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2\overline{y_1y_2}}{MF}$

$\beta_{C_{12512}}^{(1)}$

$16\pi^2\mu\frac{dC_{12512}}{d\mu}=\beta_{C_{12512}}^{(1)}$

Column	Representative term structure
T1	$\frac{ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{ig_2^2y_1y_2}{MF}$

$\beta_{C_{12515}}^{(1)}$

$16\pi^2\mu\frac{dC_{12515}}{d\mu}=\beta_{C_{12515}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2\overline{y_1y_2}}{2MF}$	$\frac{ig_2^2\overline{y_1y_2}}{2MF}$	$-\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{4MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$	$\frac{3\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{44MF}$	$\frac{45\sqrt{2}ig_2^2\overline{y_1y_2}}{16MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{12516}}^{(1)}$

$16\pi^2\mu\frac{dC_{12516}}{d\mu}=\beta_{C_{12516}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2\overline{y_1y_2}}{2MF}$	$-\frac{45\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{66}\text{lambda}12Inv2\overline{y_1}y_2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}\text{lambda}12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{66}\text{lambda}12Inv2\overline{y_1}y_2}{44MF}$	$-\frac{ig_2^2\overline{y_1}y_2}{2MF}$	$\frac{\sqrt{2}\text{lambda}12Inv1\overline{y_1}y_2}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{1267}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1267}}{d\mu}=\beta_{C_{1267}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iy e\overline{y_1}^2\overline{y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3ig_2^2\overline{y_1}^2}{8MF}$	$-\frac{iy e\overline{y_1}^2\overline{y e}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{24MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{1268}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{1268}}{d\mu}=\beta_{C_{1268}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS1Inv1\overline{y_1}^2}{4MF}$	$\frac{3g_2^2\overline{y_1}^2}{8MF}$	$-\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{8MF}$	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{12MF}$	$-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}e}{24MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{1269}}^{(1)}$

$_{16\pi^2}\mu\frac{dC_{1269}}{d\mu}=\beta_{C_{1269}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inu2\overline{y_1y_2}}{MF}$
T3	$\frac{iy_e\overline{y_1y_2y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ilambda12Inu2\overline{y_1y_2}}{6MF}$	$\frac{iy_e\overline{y_1y_2y_e}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$\frac{i\lambda 12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{i\lambda 12Inv2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i \lambda \overline{Inv1 y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$	$-\frac{3^{\frac{3}{4}} i \lambda \overline{Inv1 y_1 y_2}}{6 M F}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{12MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{12610}}^{(1)}$

$16\pi^2\mu\frac{dC_{12610}}{d\mu}=\beta_{C_{12610}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{ye\overline{y_1y_2y\overline{e}}}{8MF}$	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{6MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{12611}}^{(1)}$

$16\pi^2\mu\frac{dC_{12611}}{d\mu}=\beta_{C_{12611}}^{(1)}$

Column Representative term structure	
T1	$\frac{ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{ig_2^2y_1y_2}{MF}$

$\beta_{C_{12612}}^{(1)}$

$$16\pi^2\mu\frac{dC_{12612}}{d\mu}=\beta_{C_{12612}}^{(1)}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2\overline{y_1y_2}}{MF}$

$\beta_{C_{12615}}^{(1)}$

$16\pi^2\mu\frac{dC_{12615}}{d\mu}=\beta_{C_{12615}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}\overline{\lambda}12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^2\overline{y_1y_2}}{2MF}$	$\frac{ig_2^2\overline{y_1y_2}}{2MF}$	$-\frac{\sqrt{2}\overline{\lambda}12Inv1\overline{y_1y_2}}{4MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}g_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{66}\text{lambda}12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{2}g_1^2\overline{y_1}y_2}{16MF}$	$\frac{3\sqrt{66}\text{lambda}12Inv2\overline{y_1}y_2}{44MF}$	$\frac{45\sqrt{2}g_2^2\overline{y_1}y_2}{16MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{12616}}^{(1)}$

$16\pi^2\mu\frac{dC_{12616}}{d\mu}=\beta_{C_{12616}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2\overline{y_1y_2}}{2MF}$	$\frac{ig_2^2\overline{y_1y_2}}{2MF}$	$-\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{4MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$	$\frac{3\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{44MF}$	$\frac{45\sqrt{2}ig_2^2\overline{y_1y_2}}{16MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{12711}}^{(1)}$

$16\pi^2\mu\frac{dC_{12711}}{d\mu}=\beta_{C_{12711}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}lambda12Inv2\overline{y_1y_2}}{6MF}$	$\frac{ye\overline{y_1y_2y\bar{e}}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{12712}}^{(1)}$

$$16\pi^2\mu\frac{dC_{12712}}{d\mu}=\beta_{C_{12712}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inu2\overline{y_1y_2}}{MF}$
T3	$\frac{iy_e\overline{y_1y_2y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}ilambda12Inu2\overline{y_1y_2}}{6MF}$	$\frac{iy_e\overline{y_1y_2y_e}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$\frac{i\lambda 12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{i\lambda 12Inv2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} i \lambda_{12} \text{Inv1} \overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} i \lambda_{12} \text{Inv1} \overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T3	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$$\beta_{C_{12713}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12713}}{d\mu}=\beta_{C_{12713}}^{(1)}$$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$

$$\beta_{C_{12714}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12714}}{d\mu}=\beta_{C_{12714}}^{(1)}$$

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$

$\beta_{C_{12811}}^{(1)}$

$16\pi^2\mu\frac{dC_{12811}}{d\mu}=\beta_{C_{12811}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{iy_e\overline{y_1y_2y\bar{e}}}{MF}$
T3	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{iy_e\overline{y_1y_2y\bar{e}}}{8MF}$	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1y_2}}{6MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$\frac{i\lambda 12Inv2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{i\lambda 12Inv2y_1y_2}{3MF}$	$-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} i\lambda_{12} Inv1 \overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} i\lambda_{12} Inv1 \overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1} y_2}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1} y_2}{MF}$
T3	$\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1} y_2}{6MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1} y_2}{6MF}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y e}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{12812}}^{(1)}$

$$16\pi^2\mu\frac{dC_{12812}}{d\mu}=\beta_{C_{12812}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\overline{lambda12Inv2y_1y_2}}{MF}$
T3	$\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{\sqrt{3}\overline{lambda12Inv2y_1y_2}}{6MF}$	$\frac{ye\overline{y_1y_2y\bar{e}}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ye\overline{y_1y_2}\overline{ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{12813}}^{(1)}$

$16\pi^2\mu\frac{dC_{12813}}{d\mu}=\beta_{C_{12813}}^{(1)}$

Column Representative term structure	
T1	$\frac{ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{ig_2^2y_1y_2}{MF}$

$\beta_{C_{12814}}^{(1)}$

$16\pi^2\mu\frac{dC_{12814}}{d\mu}=\beta_{C_{12814}}^{(1)}$

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$

$\beta_{C_{12911}}^{(1)}$

$16\pi^2\mu\frac{dC_{12911}}{d\mu}=\beta_{C_{12911}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{\textit{lambdaS2Inv1}\overline{y_2}^2}{MF}$
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{\textit{lambdaS2Inv1}\overline{y_2}^2}{4MF}$	$\frac{3g_2^2\overline{y_2}^2}{8MF}$	$-\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y}\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{8MF}$	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_2}^2}{12MF}$	$-\frac{\sqrt{3}ye\overline{y_2}^2\overline{y}\overline{e}}{24MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} i g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2 \overline{y_1 y_2}}{2 M F}$	$-\frac{i g_2^2 \overline{y_1 y_2}}{2 M F}$	$-\frac{45 \sqrt{2} i g_2^2 \overline{y_1 y_2}}{16 M F}$

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$	$-\frac{3\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{44MF}$	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{4MF}$

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{12912}}^{(1)}$

$16\pi^2\mu\frac{dC_{12912}}{d\mu}=\beta_{C_{12912}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T3	$\frac{iy e\overline{y_2}^2\overline{y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{3ig_2^2\overline{y_2}^2}{8MF}$	$-\frac{iy e\overline{y_2}^2\overline{y e}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_2}^2\overline{ye}}{24MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} \lambda a_{12} \text{Inv} \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^2 \overline{y_1 y_2}}{2 M F}$	$-\frac{i g_2^2 \overline{y_1 y_2}}{2 M F}$	$-\frac{\sqrt{2} \lambda a_{12} \text{Inv} \overline{y_1 y_2}}{4 M F}$

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2}g_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{66}\text{lambda}12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{2}g_1^2\overline{y_1}y_2}{16MF}$	$\frac{3\sqrt{66}\text{lambda}12Inv2\overline{y_1}y_2}{44MF}$	$\frac{45\sqrt{2}g_2^2\overline{y_1}y_2}{16MF}$

Column	Representative term structure
T1	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{121011}}^{(1)}$

$_{16\pi^2}{}_{\mu}\frac{dC_{121011}}{d\mu}=\beta_{C_{121011}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T3	$\frac{iy e\overline{y_2}^2\overline{y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS2Inv1\overline{y_2}^2}{4MF}$	$\frac{3ig_2^2\overline{y_2}^2}{8MF}$	$-\frac{iy e\overline{y_2}^2\overline{y e}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{8MF}$	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{12MF}$	$-\frac{\sqrt{3}iye\overline{y_2}^2\overline{ye}}{24MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2 \overline{y_1 y_2}}{2 M F}$	$\frac{i g_2^2 \overline{y_1 y_2}}{2 M F}$	$-\frac{45 \sqrt{2} g_2^2 \overline{y_1 y_2}}{16 M F}$

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2}g_1^2\overline{y}_1y_2}{MF}$
T2	$\frac{\sqrt{66}\lambda b d a 12Inv2\overline{y}_1y_2}{MF}$
T3	$\frac{\sqrt{2}\lambda b d a 12Inv1\overline{y}_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{2}g_1^2\overline{y}_1y_2}{16MF}$	$-\frac{3\sqrt{66}\lambda b d a 12Inv2\overline{y}_1y_2}{44MF}$	$\frac{\sqrt{2}\lambda b d a 12Inv1\overline{y}_1y_2}{4MF}$

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{121012}}^{(1)}$

$_{16\pi^2} \mu \frac{dC_{121012}}{d\mu} = \beta_{C_{121012}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{g_2^2 \overline{y_2}^2}{MF}$
T2	$\frac{lambda S2 Inv 1 \overline{y_2}^2}{MF}$
T3	$\frac{ye \overline{y_2}^2 \overline{y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2 \overline{y_2}^2}{8MF}$	$-\frac{lambda S2 Inv 1 \overline{y_2}^2}{4MF}$	$\frac{ye \overline{y_2}^2 \overline{y e}}{8MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{8MF}$	$-\frac{\sqrt{3}\lambda\overline{y_2}^2}{12MF}$	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{24MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\frac{g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} i g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2 \overline{y_1 y_2}}{2 M F}$	$-\frac{i g_2^2 \overline{y_1 y_2}}{2 M F}$	$-\frac{45 \sqrt{2} i g_2^2 \overline{y_1 y_2}}{16 M F}$

Terms 4 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{2}ig_1^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{44MF}$	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{4MF}$

Terms 5 of 5

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{2211}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{2211}}{d\mu}=\beta_{C_{2211}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{4lambdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{2lambdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{2212}}^{(1)}$

$16\pi^2\mu\frac{dC_{2212}}{d\mu}=\beta_{C_{2212}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{i\lambda bdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{4i\lambda bdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{2\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{2i\lambda bdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{2222}}^{(1)}$

$16\pi^2\,_{\mu}\frac{dC_{2222}}{d\mu}=\beta_{C_{2222}}^{(1)}$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$
T2	$\frac{lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{4lambdaT_3\overline{y_1y_2}}{3MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\frac{2lambdaT_3\overline{y_1y_2}}{MF}$

$\beta_{C_{2257}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{2257}}{d\mu}=\beta_{C_{2257}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{2258}}^{(1)}$

$16\pi^2\mu\frac{dC_{2258}}{d\mu}=\beta_{C_{2258}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3}{3^4}\frac{ilambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3}{3^4}\frac{ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3}{3^4}\frac{ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3}{3^4}\frac{ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3}{3^4}\frac{ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3}{24}\frac{3}{3^4}\frac{ig_2^2\overline{y_1y_2}}{MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22511}}^{(1)}$

$16\pi^2\mu\frac{dC_{22511}}{d\mu}=\beta_{C_{22511}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T2	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$
T2	$\frac{3^{\frac{3}{4}} ye y_1 y_2 y e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{11 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{8MF}$	$-\frac{3^{\frac{3}{4}} ye y_1 y_2 ye}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22512}}^{(1)}$

$16\pi^2\mu\frac{dC_{22512}}{d\mu}=\beta_{C_{22512}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1} y_2 y\overline{e}}{MF}$
T2	$\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1} y_2}{MF}$
T3	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1} y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ye\overline{y_1} y_2 y\overline{e}}{12MF}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1} y_2}{6MF}$	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1} y_2}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$	$-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22513}}^{(1)}$

$16\pi^2\mu\frac{dC_{22513}}{d\mu}=\beta_{C_{22513}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$	$-\frac{49\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{66}\lambda\text{Inv}2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}\lambda\text{Inv}1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{66}\lambda\text{Inv}2\overline{y_1y_2}}{44MF}$	$\frac{\sqrt{2}\lambda\text{Inv}1\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{22514}}^{(1)}$

$16\pi^2\mu\frac{dC_{22514}}{d\mu}=\beta_{C_{22514}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2\overline{y_1y_2}}{MF}$	$-\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{49\sqrt{2}ig_2^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{2}ig_1^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{66}ilambda12Inv2\overline{y_1}y_2}{44MF}$	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1}y_2}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{2267}}^{(1)}$

$16\pi^2\mu\frac{dC_{2267}}{d\mu}=\beta_{C_{2267}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{2268}}^{(1)}$

$16\pi^2\mu\frac{dC_{2268}}{d\mu}=\beta_{C_{2268}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22611}}^{(1)}$

$16\pi^2\mu\frac{dC_{22611}}{d\mu}=\beta_{C_{22611}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y \overline{e}}}{MF}$
T2	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} i lambda 12 Inv 1 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} ye \overline{y_1 y_2 y \overline{e}}}{12MF}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}} i lambda 12 Inv 1 \overline{y_1 y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22612}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22612}}{d\mu}=\beta_{C_{22612}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$
T2	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{M F}$
T3	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\frac{3^{\frac{3}{4}} \lambda 12 I n v 1 \overline{y_1 y_2}}{6 M F}$	$\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 3

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{11 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22613}}^{(1)}$

$16\pi^2\mu\frac{dC_{22613}}{d\mu}=\beta_{C_{22613}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}i\lambda da12Inv1\overline{y_1y_2}}{4MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$	$\frac{3\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{44MF}$	$\frac{49\sqrt{2}ig_2^2\overline{y_1y_2}}{16MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{22614}}^{(1)}$

$16\pi^2\mu\frac{dC_{22614}}{d\mu}=\beta_{C_{22614}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{ig_2^2\overline{y_1y_2}}{MF}$	$-\frac{49\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{66}\lambda\text{Inv}2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}\lambda\text{Inv}1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{66}\lambda\text{Inv}2\overline{y_1y_2}}{44MF}$	$\frac{\sqrt{2}\lambda\text{Inv}1\overline{y_1y_2}}{4MF}$	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{2277}}^{(1)}$

$16\pi^2\,_{\mu}\frac{dC_{2277}}{d\mu}=\beta_{C_{2277}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{ye\overline{y_1}^2\overline{y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\overline{S_1}Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$-\frac{\sqrt{3}\lambda\overline{S_1}Inv1\overline{y_1}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{2278}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{2278}}{d\mu}=\beta_{C_{2278}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{ig_2^2\overline{y_1}^2}{MF}$
T3	$\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{ilambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3ig_2^2\overline{y_1}^2}{4MF}$	$-\frac{iy_e\overline{y_1}^2\overline{y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{2279}}^{(1)}$

$16\pi^2\mu\frac{dC_{2279}}{d\mu}=\beta_{C_{2279}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}\overline{\lambda a_{12}Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\overline{e}}}{4MF}$	$\frac{\sqrt{3}\overline{\lambda a_{12}Inv2y_1y_2}}{3MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{6MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{22710}}^{(1)}$

$16\pi^2\mu\frac{dC_{22710}}{d\mu}=\beta_{C_{22710}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{iy_e\overline{y_1y_2y\bar{e}}}{MF}$
T3	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{iy_e\overline{y_1y_2y\bar{e}}}{4MF}$	$\frac{\sqrt{3}i\lambda da12Inv2\overline{y_1y_2}}{3MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{3} i y e \overline{y_1 y_2} y e}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2 i \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3 M F}$	$-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{2 M F}$	$\frac{\sqrt{3} i y e \overline{y_1 y_2} y e}{6 M F}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{22711}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22711}}{d\mu}=\beta_{C_{22711}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{66}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$	$\frac{3\sqrt{66}lambda12Inv2\overline{y_1y_2}}{44MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{33\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}ye\overline{y_1y_2ye}}{8MF}$

$\beta_{C_{22712}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22712}}{d\mu}=\beta_{C_{22712}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{2} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{44 M F}$	$\frac{33\sqrt{2} i g_2^2 \overline{y_1 y_2}}{16 M F}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{22715}}^{(1)}$

$16\pi^2\mu\frac{dC_{22715}}{d\mu}=\beta_{C_{22715}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{2}g_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{2}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{ig_2^2y_1y_2}{MF}$	$\frac{\sqrt{2}g_2^2y_1y_2}{4MF}$	$-\frac{49\sqrt{2}ig_2^2y_1y_2}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$	$-\frac{3\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{44MF}$	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{22716}}^{(1)}$

$16\pi^2\mu\frac{dC_{22716}}{d\mu}=\beta_{C_{22716}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^2\overline{y_1y_2}}{MF}$	$-\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{66}\text{lambda}12Inv2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}\text{lambda}12Inv2\overline{y_1y_2}}{44MF}$	$\frac{49\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{2}ye\overline{y_1y_2ye}}{8MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$

$\beta_{C_{2288}}^{(1)}$

$16\pi^2\,\mu\,\frac{dC_{2288}}{d\mu}=\beta_{C_{2288}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$
T2	$\frac{g_2^2\overline{y_1}^2}{MF}$
T3	$\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$	$\frac{3g_2^2\overline{y_1}^2}{4MF}$	$-\frac{ye\overline{y_1}^2\overline{y\overline{e}}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$	$\frac{\sqrt{3}\lambda S1Inv1\overline{y_1}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y}\overline{e}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{2289}}^{(1)}$

$_{16\pi^2}\mu\frac{dC_{2289}}{d\mu}=\beta_{C_{2289}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inu2\overline{y_1y_2}}{MF}$
T3	$\frac{iy_e\overline{y_1y_2y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{3}ilambda12Inu2\overline{y_1y_2}}{3MF}$	$\frac{iy_e\overline{y_1y_2y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{ilambda12Inv2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{2ilambda12Inv2y_1y_2}{3MF}$	$\frac{\sqrt{3}ig_2^2y_1y_2}{2MF}$	$-\frac{\sqrt{3}iye\overline{y_1y_2ye}}{6MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{22810}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22810}}{d\mu}=\beta_{C_{22810}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{ye\overline{y_1y_2y\overline{e}}}{MF}$
T3	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{3g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{ye\overline{y_1y_2y\overline{e}}}{4MF}$	$\frac{\sqrt{3}\overline{\lambda da12Inv2y_1y_2}}{3MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\textit{lambda12Inv2}\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2\textit{lambda12Inv2}\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{2MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{6MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{22811}}^{(1)}$

$16\pi^2\mu\frac{dC_{22811}}{d\mu}=\beta_{C_{22811}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{2}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{2}ig_1^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}g_2^2y_1y_2}{4MF}$	$-\frac{33\sqrt{2}ig_2^2y_1y_2}{16MF}$	$-\frac{3\sqrt{2}ig_1^2y_1y_2}{16MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{2}i\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2}i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{44 M F}$	$\frac{\sqrt{2}i\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{4 M F}$	$\frac{\sqrt{2}i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{22812}}^{(1)}$

$16\pi^2\mu\frac{dC_{22812}}{d\mu}=\beta_{C_{22812}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{66}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$	$\frac{3\sqrt{66}lambda12Inv2\overline{y_1y_2}}{44MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{33\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}ye\overline{y_1y_2ye}}{8MF}$

$\beta_{C_{22815}}^{(1)}$

$16\pi^2\mu\frac{dC_{22815}}{d\mu}=\beta_{C_{22815}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^2\overline{y_1y_2}}{MF}$	$-\frac{49\sqrt{2}g_2^2\overline{y_1y_2}}{16MF}$	$-\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{66}\lambda\text{Inv}2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}\lambda\text{Inv}1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{66}\lambda\text{Inv}2\overline{y_1y_2}}{44MF}$	$\frac{\sqrt{2}\lambda\text{Inv}1\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}ig_2^2\overline{y_1y_2}}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{22816}}^{(1)}$

$16\pi^2\mu\frac{dC_{22816}}{d\mu}=\beta_{C_{22816}}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{ig_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{2}g_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{2}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{ig_2^2y_1y_2}{MF}$	$\frac{\sqrt{2}g_2^2y_1y_2}{4MF}$	$-\frac{49\sqrt{2}ig_2^2y_1y_2}{16MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$	$-\frac{3\sqrt{66}ilambda12Inv2\overline{y_1y_2}}{44MF}$	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{4MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{2299}}^{(1)}$

$16\pi^2\mu\frac{dC_{2299}}{d\mu}=\beta_{C_{2299}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{ye\overline{y_2}^2\overline{y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\overline{S_2}Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}\lambda\overline{S_2}Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{22910}}^{(1)}$

$16\pi^2\mu\frac{dC_{22910}}{d\mu}=\beta_{C_{22910}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{iy_e\overline{y_2}^2\overline{y_e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{ilambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{iy_e\overline{y_2}^2\overline{y_e}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{4MF}$	$-\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{6MF}$	$\frac{\sqrt{3}iye\overline{y_2}^2\overline{ye}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{22911}}^{(1)}$

$16\pi^2\mu\frac{dC_{22911}}{d\mu}=\beta_{C_{22911}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 ye}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22912}}^{(1)}$

$$16\pi^2\mu\frac{dC_{22912}}{d\mu}=\beta_{C_{22912}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{22913}}^{(1)}$

$16\pi^2\mu\frac{dC_{22913}}{d\mu}=\beta_{C_{22913}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}i\lambda a_{12}Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}i\lambda a_{12}Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{2} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{44 M F}$	$\frac{33\sqrt{2} i g_2^2 \overline{y_1 y_2}}{16 M F}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$

$\beta_{C_{22914}}^{(1)}$

$16\pi^2\mu\frac{dC_{22914}}{d\mu}=\beta_{C_{22914}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{2}g_1^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{66}\lambda\overline{12Inv2\overline{y_1}y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33\sqrt{2}g_2^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{2}g_1^2\overline{y_1}y_2}{16MF}$	$-\frac{3\sqrt{66}\lambda\overline{12Inv2\overline{y_1}y_2}}{44MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}lambda12Inv1y_1y_2}{MF}$
T2	$\frac{\sqrt{2}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}lambda12Inv1y_1y_2}{4MF}$	$-\frac{\sqrt{2}ig_2^2y_1y_2}{4MF}$	$\frac{\sqrt{2}ye\overline{y}_1y_2\overline{y}e}{8MF}$

$\beta_{C_{221010}}^{(1)}$

$_{16\pi^2}^2\mu\frac{dC_{221010}}{d\mu}=\beta_{C_{221010}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$
T2	$\frac{g_2^2\overline{y_2}^2}{MF}$
T3	$\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$	$\frac{3g_2^2\overline{y_2}^2}{4MF}$	$-\frac{ye\overline{y_2}^2\overline{y\overline{e}}}{4MF}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$
T2	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_2}^2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_2}^2\overline{y}\overline{e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_2}^2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_2}^2\overline{y}\overline{e}}{12MF}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{C_{221011}}^{(1)}$

$_{16\pi^2} \mu \frac{dC_{221011}}{d\mu} = \beta_{C_{221011}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3}{3^4} \frac{g_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3}{3^4} \frac{lambda12Inv1 \overline{y_1 y_2}}{MF}$
T3	$\frac{3}{3^4} \frac{g_1^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{17 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} lambda12Inv1 \overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{3MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{221012}}^{(1)}$

$16\pi^2\,_{\mu}\frac{dC_{221012}}{d\mu}=\beta_{C_{221012}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$	$\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y e}}{12MF}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{C_{221013}}^{(1)}$

$_{16\pi^2}^{\mu}\frac{dC_{221013}}{d\mu}=\beta_{C_{221013}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{66}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}lambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}g_1^2\overline{y_1y_2}}{16MF}$	$\frac{3\sqrt{66}lambda12Inv2\overline{y_1y_2}}{44MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{2}ye\overline{y_1}y_2ye}{MF}$
T3	$\frac{\sqrt{2}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{33\sqrt{2}g_2^2\overline{y_1}y_2}{16MF}$	$-\frac{\sqrt{2}ye\overline{y_1}y_2ye}{8MF}$	$\frac{\sqrt{2}ig_2^2\overline{y_1}y_2}{4MF}$

$\beta_{C_{221014}}^{(1)}$

$_{16\pi^2}^{\mu}\frac{dC_{221014}}{d\mu}=\beta_{C_{221014}}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{2}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2}g_2^2\overline{y_1y_2}}{4MF}$	$-\frac{\sqrt{2}ilambda12Inv1\overline{y_1y_2}}{4MF}$	$\frac{3\sqrt{2}ig_1^2\overline{y_1y_2}}{16MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$
T2	$\frac{\sqrt{2} i g_2^2 \overline{y_1 y_2}}{M F}$
T3	$\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{66}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{44 M F}$	$\frac{33\sqrt{2} i g_2^2 \overline{y_1 y_2}}{16 M F}$	$-\frac{\sqrt{2} i y e \overline{y_1 y_2 y e}}{8 M F}$